

```
#define BLYNK_TEMPLATE_ID "TMPL6cApWdZfH"
#define BLYNK_TEMPLATE_NAME "Smart Greenhouse"
#define BLYNK_AUTH_TOKEN "GhwnqULnRj_T4gF9_In5eUECUgEk9vIO"
```

```
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <DHT.h>
#include <ESP32Servo.h>
#include <Wire.h>
```

```
// =====
// WiFi
// =====
char ssid[] = "Austian";
char pass[] = "austMFAAZAP";
```

```
// =====
// PIN DEFINITIONS
// =====
#define DHT_PIN    4
#define SOIL_PIN   34
#define RAIN_PIN   32
#define LDR_PIN    35
```

```
#define SERVO_PIN  13

#define PUMP_RELAY 26
#define LIGHT_RELAY 25
#define FAN_RELAY  33
```

```
#define LCD_SDA    21
```

```
#define LCD_SCL    22
```

```
#define LCD_ADDR   0x27
```

```
// =====
```

```
// SENSOR
```

```
// =====
```

```
#define DHTTYPE DHT22
```

```
DHT dht(DHT_PIN, DHTTYPE);
```

```
Servo roofServo;
```

```
BlynkTimer timer;
```

```
// =====
```

```
// VARIABLES
```

```
// =====
```

```
bool autoMode = true;
```

```
bool pumpState = false;
```

```
bool fanState = false;
```

```
bool lightState = false;
```

```
bool roofClosed = false;
```

```
float temperature = 0;
```

```
float humidity = 0;
```

```

int soilValue = 0;

int rainValue = 0;

int ldrValue = 0;


// =====

// THRESHOLDS

// =====


// Soil
#define SOIL_PUMP_ON 3000
#define SOIL_PUMP_OFF 2500


// Rain
#define RAIN_THRESHOLD 3500


// Fan
#define FAN_ON_TEMP 30.0
#define FAN_OFF_TEMP 28.0


// =====

// LCD LOW LEVEL DRIVER

// =====


void lcdWrite4Bits(byte data)
{
    Wire.beginTransmission(LCD_ADDR);

    Wire.write(data | 0x08);

    Wire.endTransmission();

```

```
Wire.beginTransaction(LCD_ADDR);  
Wire.write(data | 0x0C);  
Wire.endTransmission();
```

```
Wire.beginTransaction(LCD_ADDR);  
Wire.write(data | 0x08);  
Wire.endTransmission();  
}
```

```
void lcdCommand(byte cmd)  
{  
    lcdWrite4Bits(cmd & 0xF0);  
    lcdWrite4Bits((cmd << 4) & 0xF0);  
}
```

```
void lcdData(byte data)  
{  
    lcdWrite4Bits((data & 0xF0) | 0x01);  
    lcdWrite4Bits(((data << 4) & 0xF0) | 0x01);  
}
```

```
void lcdClear()  
{  
    lcdCommand(0x01);  
    delay(2);  
}
```

```
void lcdSetCursor(byte col, byte row)  
{
```

```
byte address;
```

```
if (row == 0)
```

```
    address = 0x80 + col;
```

```
else
```

```
    address = 0xC0 + col;
```

```
    lcdCommand(address);
```

```
}
```

```
void lcdPrint(String text)
```

```
{
```

```
    for (int i = 0; i < text.length(); i++)
```

```
    {
```

```
        lcdData(text[i]);
```

```
    }
```

```
}
```

```
void lcdPrintLine(byte row, String text)
```

```
{
```

```
    lcdSetCursor(0, row);
```

```
    while (text.length() < 16)
```

```
    {
```

```
        text += " ";
```

```
    }
```

```
    if (text.length() > 16)
```

```
    {
```

```
    text = text.substring(0, 16);  
}
```

```
    lcdPrint(text);  
}
```

```
void lcdInit()  
{  
    Wire.begin(LCD_SDA, LCD_SCL);
```

```
    delay(50);
```

```
    lcdWrite4Bits(0x30);  
    delay(5);
```

```
    lcdWrite4Bits(0x30);  
    delay(1);
```

```
    lcdWrite4Bits(0x30);  
    delay(1);
```

```
    lcdWrite4Bits(0x20);  
    delay(1);
```

```
    lcdCommand(0x28);  
    lcdCommand(0x0C);  
    lcdCommand(0x01);
```

```
    delay(2);
```

```

    lcdCommand(0x06);
}

// =====
// RELAY CONTROL
// LOW = ON
// HIGH = OFF
// =====

void setPump(bool state)
{
    pumpState = state;

    if (state)
        digitalWrite(PUMP_RELAY, LOW);
    else
        digitalWrite(PUMP_RELAY, HIGH);
}

void setFan(bool state)
{
    fanState = state;

    if (state)
        digitalWrite(FAN_RELAY, LOW);
    else
        digitalWrite(FAN_RELAY, HIGH);
}

```

```

void setLight(bool state)
{
    lightState = state;

    if (state)
        digitalWrite(LIGHT_RELAY, LOW);
    else
        digitalWrite(LIGHT_RELAY, HIGH);
}

// =====
// ROOF CONTROL
// =====

void closeRoof()
{
    roofServo.write(0);
    roofClosed = true;
}

void openRoof()
{
    roofServo.write(90);
    roofClosed = false;
}

// =====
// BLYNK AUTO / MANUAL

```

```
// V0
// 1 = AUTO
// 0 = MANUAL
// =====

BLYNK_WRITE(V0)
{
    int value = param.asInt();

    if (value == 1)
        autoMode = true;
    else
        autoMode = false;

    Serial.print("Mode: ");

    if (autoMode)
        Serial.println("AUTO");
    else
        Serial.println("MANUAL");
}

// =====
// MANUAL PUMP V12
// =====

BLYNK_WRITE(V12)
{
    if (!autoMode)
```

```
{  
    setPump(param.asInt());  
}  
}
```

```
// =====  
// MANUAL FAN V13  
// =====
```

```
BLYNK_WRITE(V13)  
{  
    if (!autoMode)  
    {  
        setFan(param.asInt());  
    }  
}
```

```
// =====  
// MANUAL LIGHT V14  
// =====
```

```
BLYNK_WRITE(V14)  
{  
    if (!autoMode)  
    {  
        setLight(param.asInt());  
    }  
}
```

```
// =====  
// MANUAL ROOF V15  
// 1 = CLOSE  
// 0 = OPEN  
// =====
```

```
BLYNK_WRITE(V15)  
{  
  if (!autoMode)  
  {  
    if (param.asInt() == 1)  
      closeRoof();  
    else  
      openRoof();  
  }  
}
```

```
// =====  
// SENSOR READING + AUTO CONTROL  
// =====
```

```
void readSensors()  
{  
  // =====  
  // DHT22  
  // =====
```

```
float newTemp = dht.readTemperature();  
float newHum = dht.readHumidity();
```

```
if (!isnan(newTemp))
```

```
    temperature = newTemp;
```

```
if (!isnan(newHum))
```

```
    humidity = newHum;
```

```
// =====
```

```
// SOIL
```

```
// =====
```

```
soilValue = analogRead(SOIL_PIN);
```

```
// =====
```

```
// RAIN
```

```
// =====
```

```
rainValue = analogRead(RAIN_PIN);
```

```
// =====
```

```
// LDR
```

```
// LOW = BRIGHT
```

```
// HIGH = DARK
```

```
// =====
```

```
// Direct reading:
```

```
// Dark -> HIGH
```

```
// Bright -> LOW
```

```
ldrValue = digitalRead(LDR_PIN);
```

```
// =====
```

```
// AUTO MODE
```

```
// =====
```

```
if (autoMode)
```

```
{
```

```
    // -----
```

```
    // SOIL → PUMP
```

```
    // -----
```

```
    if (soilValue >= SOIL_PUMP_ON)
```

```
    {
```

```
        setPump(true);
```

```
    }
```

```
    else if (soilValue <= SOIL_PUMP_OFF)
```

```
    {
```

```
        setPump(false);
```

```
    }
```

```
    // -----
```

```
    // TEMPERATURE → FAN
```

```
    // -----
```

```
    if (!isnan(newTemp))
```

```
    {
```

```
        if (temperature >= FAN_ON_TEMP)
```

```
        {
```

```
    setFan(true);  
}  
else if (temperature <= FAN_OFF_TEMP)  
{  
    setFan(false);  
}  
}
```

```
// -----  
// LDR → LIGHT  
// HIGH = DARK  
// LOW = BRIGHT  
// -----
```

```
if (ldrValue == 1) {  
    setLight(true);  
}  
else {  
    setLight(false);  
}
```

```
// -----  
// RAIN → ROOF  
// -----
```

```
if (rainValue < RAIN_THRESHOLD)  
{  
    closeRoof();  
}
```

```
else
{
    openRoof();
}
}

// =====
// SERIAL MONITOR
// =====

Serial.println("-----");

Serial.print("Temperature: ");
Serial.print(temperature);
Serial.println(" C");

Serial.print("Humidity: ");
Serial.print(humidity);
Serial.println(" %");

Serial.print("Soil: ");
Serial.println(soilValue);

Serial.print("Rain: ");
Serial.println(rainValue);

Serial.print("LDR: ");

if (ldrValue == HIGH)
```

```
    Serial.println("DARK");
else
    Serial.println("BRIGHT");

Serial.print("Pump: ");
Serial.println(pumpState ? "ON" : "OFF");

Serial.print("Fan: ");
Serial.println(fanState ? "ON" : "OFF");

Serial.print("Light: ");
Serial.println(lightState ? "ON" : "OFF");

Serial.print("Roof: ");
Serial.println(roofClosed ? "CLOSED" : "OPEN");

Serial.print("Mode: ");
Serial.println(autoMode ? "AUTO" : "MANUAL");
}

// =====
// SEND DATA TO BLYNK
// =====

void sendToBlynk()
{
    Blynk.virtualWrite(V2, temperature);
    Blynk.virtualWrite(V3, humidity);
```

```

Blynk.virtualWrite(V4, soilValue);
Blynk.virtualWrite(V5, rainValue);
Blynk.virtualWrite(V6, ldrValue);

Blynk.virtualWrite(V7, pumpState);
Blynk.virtualWrite(V8, fanState);
Blynk.virtualWrite(V9, lightState);

Blynk.virtualWrite(V10, roofClosed);
Blynk.virtualWrite(V11, autoMode);

Blynk.virtualWrite(V12, pumpState);
Blynk.virtualWrite(V13, fanState);
Blynk.virtualWrite(V14, lightState);
Blynk.virtualWrite(V15, roofClosed);
}

// =====
// LCD DISPLAY
// =====

void updateLCD()
{
  // Page 1

  lcdPrintLine(
    0,
    "T:" + String(temperature, 1) +
    "C H:" + String(humidity, 0) + "%"

```

```
);
```

```
lcdPrintLine(  
    1,  
    "Soil:" + String(soilValue)  
);
```

```
delay(1500);
```

```
// Page 2
```

```
lcdPrintLine(  
    0,  
    "Rain:" + String(rainValue)  
);
```

```
lcdPrintLine(  
    1,  
    "L:" + String(lightState ? "ON " : "OFF") +  
    " F:" + String(fanState ? "ON " : "OFF")  
);
```

```
delay(1500);
```

```
// Page 3
```

```
lcdPrintLine(  
    0,  
    "Pump:" + String(pumpState ? "ON " : "OFF")
```

```
);
```

```
  lcdPrintLine(
```

```
    1,
```

```
    "Roof:" + String(roofClosed ? "CLOSED" : "OPEN")
```

```
  );
```

```
}
```

```
// =====
```

```
// BLYNK CONNECTED
```

```
// =====
```

```
BLYNK_CONNECTED()
```

```
{
```

```
  Serial.println("Blynk Connected!");
```

```
  Blynk.syncVirtual(V0);
```

```
  Blynk.syncVirtual(V12);
```

```
  Blynk.syncVirtual(V13);
```

```
  Blynk.syncVirtual(V14);
```

```
  Blynk.syncVirtual(V15);
```

```
}
```

```
// =====
```

```
// SETUP
```

```
// =====
```

```
void setup()
```

```
{  
  Serial.begin(115200);  
  
  // Relay pins  
  pinMode(PUMP_RELAY, OUTPUT);  
  pinMode(LIGHT_RELAY, OUTPUT);  
  pinMode(FAN_RELAY, OUTPUT);  
  
  // All relays OFF at startup  
  digitalWrite(PUMP_RELAY, HIGH);  
  digitalWrite(LIGHT_RELAY, HIGH);  
  digitalWrite(FAN_RELAY, HIGH);  
  
  // LDR  
  pinMode(LDR_PIN, INPUT);  
  
  // DHT  
  dht.begin();  
  
  // Servo  
  roofServo.setPeriodHertz(50);  
  
  roofServo.attach(  
    SERVO_PIN,  
    500,  
    2400  
  );  
  
  // Start with roof open
```

```
openRoof();

// LCD
lcdInit();

lcdPrintLine(0, "SMART GREENHOUSE");
lcdPrintLine(1, "Starting...");

delay(2000);

// Blynk
Blynk.begin(
  BLYNK_AUTH_TOKEN,
  ssid,
  pass
);

// Timers
timer.setInterval(2000L, readSensors);
timer.setInterval(3000L, sendToBlynk);
timer.setInterval(5000L, updateLCD);

lcdClear();

Serial.println();
Serial.println("=====");
Serial.println(" SMART GREENHOUSE STARTED");
Serial.println("=====");
}
```

```
// =====
```

```
// LOOP
```

```
// =====
```

```
void loop()
```

```
{
```

```
  Blynk.run();
```

```
  timer.run();
```

```
}
```